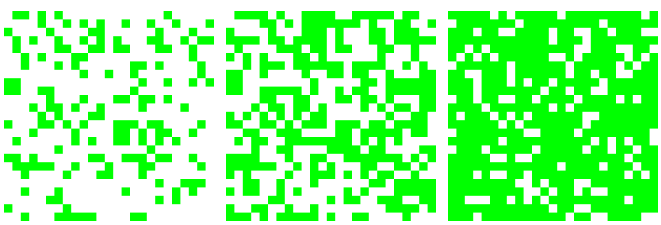
# Exercise 2: Percolation

Francesco Miranda Mendes*

This work focused on the implementation of canonical percolation configurations, the study of the burning method, and, finally, on a statistical analysis comparing percolation lattices for various parameters.

## I. GENERATING CANONICAL PERCOLATION CONFIGURATIONS

We began by generating $L \times L$ lattices, where $L$ is the number of cells along one side of a square. The configurations are canonical, that is, the occupation probability of a cell is constant. To create the network, the generator `drand48()` was used, which generates $x \in [0, 1]$ randomly for each cell. If $x \leq p$, where $p$ is the occupation probability, the cell becomes occupied. Otherwise, it remains empty. Therefore, there are $2^{L \times L}$ possible lattice configurations.



(a) $p = 0.25$ (b) $p = 0.50$ (c) $p = 0.70$

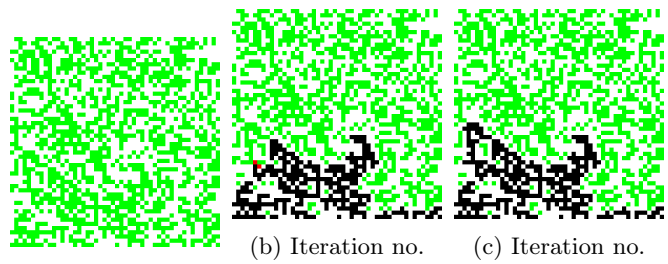
Figure 1: Canonical percolation configurations for $L = 25$ and different values of $p$, obtained using *latticeview.h*. Occupied cell: **green**; Empty: **white**.

Observing Figure 1, one notices that the only configuration in which percolation occurs, that is, in which there is at least one path of adjacent occupied cells starting in the first row and ending in the last row (25th), is the one shown in panel (c).

## II. BURNING METHOD

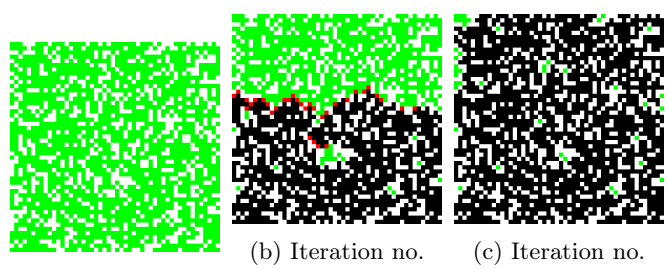
Next, the burning method was developed for these configurations. As the name suggests, this method consists of simulating a fire, as well as its propagation throughout space, which in this case is a square lattice of side $L$. For this purpose, the following algorithm was implemented:

1. Ignite the occupied cells in the first row: the cells become red;
2. Check whether horizontally and vertically adjacent cells (ignoring boundaries) are occupied;
3. If the neighboring cells are occupied, the fire propagates and they become red. The previous cell becomes black to indicate that it has burned (there is no longer propagation through that cell);
4. Continue iterating until there is a cell without occupied neighbors, or until the last row is reached (percolating case).



(a) Initial config. (b) Iteration no. 38. (c) Iteration no. 57.

Figure 2: Iterative evolution of the canonical configuration for $L = 75$ and $p = 0.5$. Panel (c) represents the end of the fire.



(a) Initial config. (b) Iteration no. 38. (c) Iteration no. 72.

Figure 3: Progress of the Burning Method for $L = 75$ and $p = 0.7$. Burning cell: **Red**; Burned cell: **Black**.

From Figures 2 and 3, it can be seen that, for $p = 0.5$, the fire does not percolate. For $p = 0.7$, however, there is a percolating cluster (the fire reached the top). For both cases, the duration time of the fire was calculated: $t_i(p = 0.50) = 57$ and $t_i(p = 0.70) = 76$, which, by defining 1 iteration as the unit of time, is equivalent to the total number of iterations. The value of the shortest path length for percolating clusters was also obtained, $C_c(p = 0.70) = 56$, defined as the number of time steps

* fc59899@alunos.fc.ul.pt

required for the fire to reach the other side of the system. $C_c(p = 0.5) = 0$, since the configuration does not percolate. The burning method is intuitive and effective for illustrating percolation, but it presents limitations in efficiency and generalization: it is computationally expensive, restricted to simple geometries, and only identifies clusters connected to the initial boundary, thus not allowing complete statistical characterizations.

## III. STATISTICS

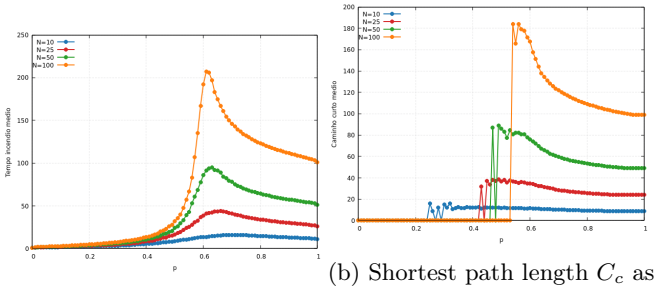
Finally, in order to draw statistically relevant conclusions, $10^3$ samples were used for each pair $(L, p)$, which guarantees a statistical error $< 3\%$, with different seeds for each sample, using the `drand48()` generator. The seeds are created in sequence, which means that, in the code, `sdrand48()` was initialized only once, so the random generator runs without being reseeded for each sample, which saves considerable computational power. Decorrelation is guaranteed, since each subsequent sample uses the next state in the pseudo-random sequence.

The average of the previously defined parameters ($t_i$, $C_c$) was evaluated. Additionally, $F_p$ is defined as the fraction of generated configurations that percolate.

| $N_{samples}$ | $p$ | $L$ | $< t_i >$ | $< C_c >$ | $F_p$ |
|---|---|---|---|---|---|
| 1000 | 0.6 | 100 | 187.303 | 165.921 | 0.647 |

Table I: Results obtained with `sdrand48(38)` as the initial seed.

Next, we investigated what happens when varying $p$ and $L$, for the same number of samples. The previous evaluation was repeated, but varying $p \in (0, 1)$ in increments of 0.01, and for $L = \{10, 25, 50, 100\}$.




(a) Average fire duration as a function of $p$ and $L$ ($N = L$).

(b) Shortest path length $C_c$ as a function of $p$ and $L$ ($N = L$).

Figure 4: For each pair $(L, p)$, 1000 samples were generated with different *seeds*.

In panel (a), one can see that the average fire duration ($< t_i >$), for different values of $L$, grows smoothly, which is explained by the increasing number of occupied cells that participate in the fire, so more iterations are needed for the fire to end. At the point where $p \approx 0.6$, $t_i$ increases abruptly. From there on, it decreases until it stabilizes. It is also noticeable that for larger $L$, the fire duration is always greater, which is consistent with the fact that there is a larger total number of cells to analyze $(L \times L)$, requiring more iterations.

From panel (b), it is clear that the graph of the average shortest path length ($< C_c >$) also shows a sharp change, which makes sense, since for systems that do not percolate $C_c = 0$, so the rapid change is due to the emergence of percolating configurations. Once again, for larger $L$, $C_c$ is generally greater because lattices with more cells have a finer scale division, which increases the value of $C_c$. In addition, it is worth noting that the probability at which the peak occurs increases with increasing $L$.

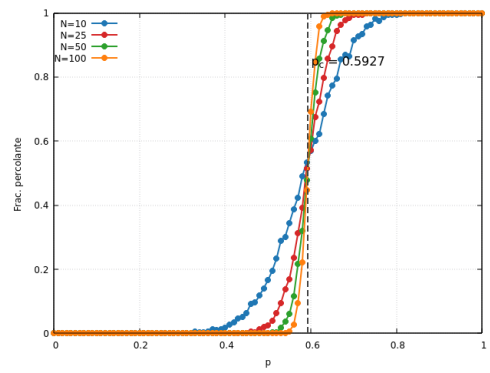



Figure 5: Fraction of samples with a percolating cluster, $F_p$, as a function of the occupation probability, $p$, for different values of $L$. The intersection point between the curves, $p_c$, is the critical probability. Results obtained for 1000 samples for each $L$.

Observing Figure 5, it is evident that the fraction of configurations that percolate is very low for low values of $p$, and as the probability gradually increases, it remains stable until what appears to be a phase transition occurs: a sharp increase in $F_p$ is observed from $p \approx 0.6$ onward. This makes sense, since with a low occupation probability there will be too few occupied cells for the fire to propagate and reach the end of the lattice. It is also noticeable that for larger $L$, the change is more abrupt. This is explained by the approach to the thermodynamic limit ($L \to \infty$), where the critical probability has been established (tabulated) as $p_c \approx 0.592746$. Revisiting what was said about the panels in Fig. 4, we conclude that, indeed, the value of $p$ at which the observed phase transitions occur, in both cases, approaches $p_c$ as $L$ increases (thermodynamic limit). The results reproduce the expected critical behavior: a percolation fraction that increases sharply around $p_c \approx 0.592746$, with a clearer transition as the size $L$ increases. The obtained value of $p_c$ and the behavior of the averages confirm the universality of the percolation transition in 2D, where macroscopic proper-

ties emerge from local fluctuations.